

Supplementary Material for “Automated Selection-based Mixture-of-Experts with Dual-stage Input–Target Pattern Learning for Sequential Recommendation”

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1 Additional Results of Empirical Studies on Input–target Patterns

This section presents the results of empirical studies on the Toys and Yelp datasets. Specifically, Table 1 reports results on the full set and individual subsets, while Table 2 reports results on the full set and augmented sets. The observations are consistent with those in the main paper.

Table 1: SASRec performance (NDCG@10) on test subsets with the configurations of the whole training set or single subsets on Toys and Yelp.

Training Configuration	$\mathcal{P}_1^{te}$	Toys $\mathcal{P}_2^{te}$	$\mathcal{P}_3^{te}$	$\mathcal{P}_1^{te}$	Yelp $\mathcal{P}_2^{te}$	$\mathcal{P}_3^{te}$
$\mathcal{D}^{tr}$	0.0134	0.0383	0.0890	0.0115	0.0183	0.1260
$\mathcal{P}_1^{tr}$	0.0142↑	0.0070	0.0074	0.0132↑	0.0073	0.0091
$\mathcal{P}_2^{tr}$	0.0032	0.0434↑	0.0107	0.0043	0.0197↑	0.0110
$\mathcal{P}_3^{tr}$	0.0033	0.0132	0.1019↑	0.0025	0.0073	0.1328↑

The **boldfaced** values denote the maximum results in each column, which are also the best performance across different test subsets of the same dataset. The symbol ↑ indicates the improvement relative to the results of the training configuration $\mathcal{D}^{tr}$ in the same column.

Table 2: SASRec performance (NDCG@10) on test subsets with the configurations of the whole or augmented training sets on Toys and Yelp.

Training Configuration	$\mathcal{P}_1^{te}$	Toys $\mathcal{P}_2^{te}$	$\mathcal{P}_3^{te}$	$\mathcal{P}_1^{te}$	Yelp $\mathcal{P}_2^{te}$	$\mathcal{P}_3^{te}$
$\mathcal{D}^{tr}$	0.0134	0.0383	0.0890	0.0115	0.0183	0.1260
$\mathcal{D}^{tr}$ and $\mathcal{P}_1^{tr}$	0.0142↑	0.0284↓	0.0693↓	0.0130↑	0.0149↓	0.1195↓
$\mathcal{D}^{tr}$ and $\mathcal{P}_2^{tr}$	0.0120↓	0.0470↑	0.0812↓	0.0090↓	0.0194↑	0.1196↓
$\mathcal{D}^{tr}$ and $\mathcal{P}_3^{tr}$	0.0110↓	0.0321↓	0.1020↑	0.0095↓	0.0166↓	0.1323↑

The symbols ↑ and ↓ indicate the relative change compared to the results of the training configuration $\mathcal{D}^{tr}$ in the same column.

2 Additional Results for In-depth Analyses of ASMoE

2.1 Impact of the Regularization Weight λ on Toys and Yelp

Figure 1 reports the sensitivity of our ASMoE to λ on the Toys and Yelp datasets. The overall trends are consistent with those in the main paper.

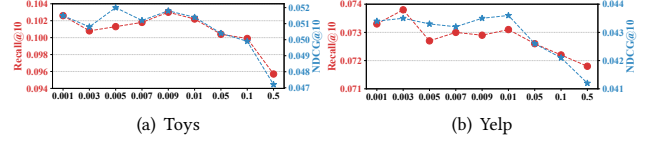


Figure 1: The performance of our ASMoE with different values of λ on Toys and Yelp. Best viewed in color.

2.2 The Generalization of our ASMoE across Dependency Levels on Toys and Yelp

Table 3 reports the performance comparison between SASRec and our ASMoE under different dependency levels on the Toys and Yelp datasets. These results again support the generalization advantage of our ASMoE.

Table 3: Performance comparison (NDCG@10) between SASRec and our ASMoE on different test subsets.

Dataset	Toys			Yelp		
Test Subset	$\mathcal{P}_1^{te}$	$\mathcal{P}_2^{te}$	$\mathcal{P}_3^{te}$	$\mathcal{P}_1^{te}$	$\mathcal{P}_2^{te}$	$\mathcal{P}_3^{te}$
SASRec	0.0134	0.0383	0.0890	0.0115	0.0183	0.126
ASMoE	0.0135	0.0420	0.1080	0.0130	0.0227	0.1353
Improv.	0.75%	9.66%	21.35%	13.04%	24.04%	7.38%